

Advanced Statistics & probability

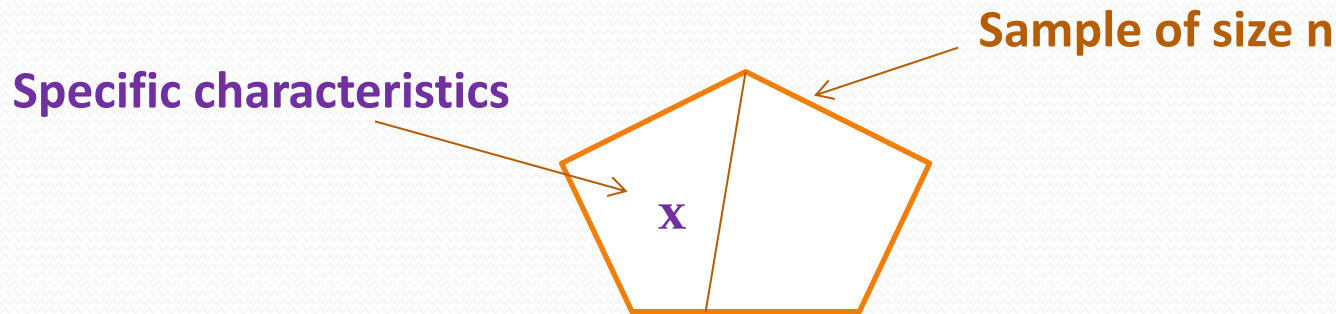
Lecture 5

By

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Interval estimation form the population proportion (P)

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population then:



The proportion is defined as:

$$\hat{P} = \frac{X}{n}, \quad 0 \leq \hat{P} \leq 1.$$

Remark:

- ✓ $P \rightarrow$ proportion of the population.
- ✓ $\hat{P} \rightarrow$ proportion of the sample.

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with proportion then:

$$Z = \frac{\hat{P} - P}{\sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}} \sim N(0, 1)$$

We note that :

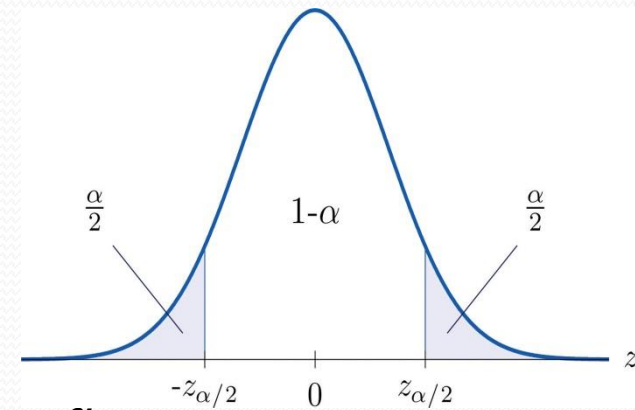
$$P\left(-z_{\frac{\alpha}{2}} < Z < z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

Which implies

$$P\left(-z_{\frac{\alpha}{2}} < \frac{\hat{P} - P}{\sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}} < z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} < \hat{P} - P < z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}\right) = 1 - \alpha$$

$$P\left(-\hat{P} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} < -P < -\hat{P} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}\right) = 1 - \alpha$$



$$P\left(\hat{P} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}(1-\hat{P})}{n}} > P > \hat{P} - z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}(1-\hat{P})}{n}}\right) = 1 - \alpha$$

Which implies finally that $(1 - \alpha)100\%$ confidence interval for the population proportion is given in the form

$$\hat{P} - z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}(1-\hat{P})}{n}} < P < \hat{P} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}(1-\hat{P})}{n}}$$

Remark: The given data of the problems are:

The proportion of the
sample $\hat{P}$

The size of the sample
 n

- Example:

A random sample of **100** patient is selected and treated by a new drug for AIDS. After six weeks **0.24** of them signs of improvement. Find **95%** confidence limits for the true proportion of all patient treatment by the new drug.

Solution

$$n = 100, \quad \hat{P} = 0.24$$

$$\alpha = 1 - 0.95 \rightarrow \alpha = 0.05 \rightarrow \frac{\alpha}{2} = 0.025 \rightarrow Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$$

95% confidence interval for proportion

$$\rightarrow \hat{P} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} < P < \hat{P} + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}$$

$$\rightarrow 0.24 - 1.96 * \sqrt{\frac{0.24(1 - 0.24)}{100}} < P < 0.24 + 1.96 * \sqrt{\frac{0.24(1 - 0.24)}{100}}$$
$$\rightarrow 0.1563 < P < 0.3237$$

- Example:

In a random sample of **500** families owing television sets in the city of Hamilton, Canada, it was found that **340** owned colors sets. Find **95%** confidence interval for the actual proportion of families in this city with color sets.

Solution

$$n = 500, \quad X = 340, \quad \hat{P} = \frac{340}{500} = 0.68$$

$$\alpha = 1 - 0.95 \rightarrow \alpha = 0.05 \rightarrow \frac{\alpha}{2} = 0.025 \rightarrow Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$$

95% confidence interval for proportion

$$\begin{aligned} &\rightarrow \hat{P} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} < P < \hat{P} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} \\ &\rightarrow 0.68 - 1.96 * \sqrt{\frac{0.68(1 - 0.68)}{500}} < P < 0.68 + 1.96 * \sqrt{\frac{0.68(1 - 0.68)}{500}} \\ &\rightarrow 0.64 < P < 0.72 \end{aligned}$$

- Example:

A national survey was run to estimate the proportion P of individuals 16 years old and under who smoke regularly. Out of **1000** individual interviewed **200** smoked regularly. Find **95%** confidence interval for P .

Solution

$$n = 1000, \quad X = 200, \quad \hat{P} = \frac{200}{1000} = 0.2$$

$$\alpha = 1 - 0.95 \rightarrow \alpha = 0.05 \rightarrow \frac{\alpha}{2} = 0.025 \rightarrow Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$$

95% confidence interval for proportion

$$\rightarrow \hat{P} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}} < P < \hat{P} + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}(1 - \hat{P})}{n}}$$

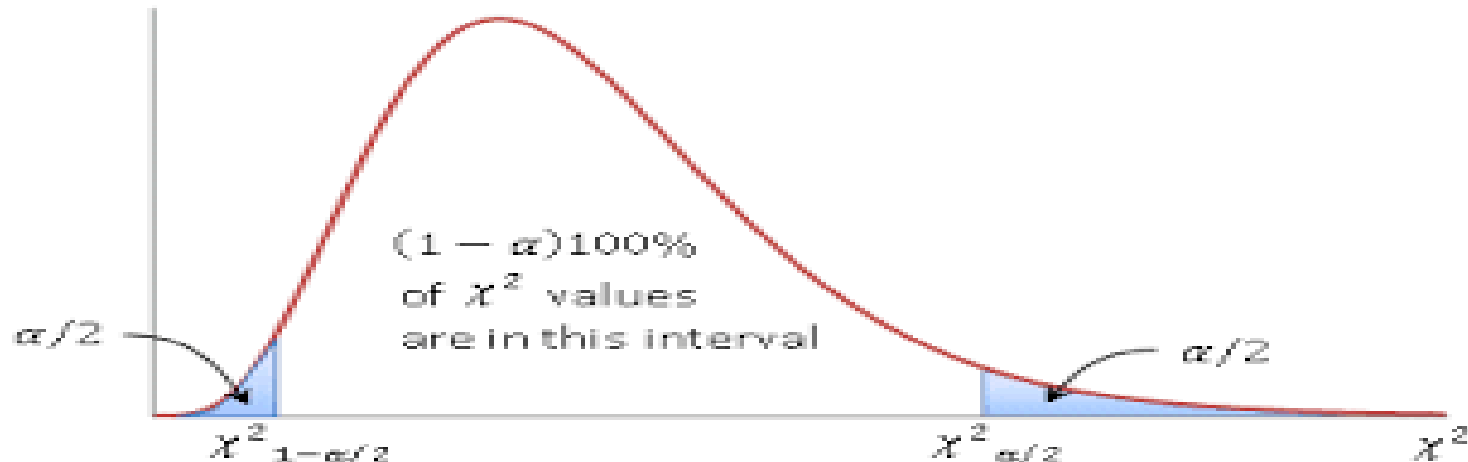
$$\rightarrow 0.2 - 1.96 * \sqrt{\frac{0.2(1 - 0.2)}{1000}} < P < 0.2 + 1.96 * \sqrt{\frac{0.2(1 - 0.2)}{1000}}$$

$$\rightarrow 0.175 < P < 0.225$$

Interval estimation form the population variance (σ^2)

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from normal population with mean μ and variance σ^2 then:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$



We note that :

$$P\left(\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)} < Z < \chi^2_{\left(\frac{\alpha}{2}, n-1\right)}\right) = 1 - \alpha$$

Which implies

$$P\left(\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\left(\frac{\alpha}{2}, n-1\right)}\right) = 1 - \alpha$$

$$P\left(\frac{1}{\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)}} > \frac{\sigma^2}{(n-1)s^2} > \frac{1}{\chi^2_{\left(\frac{\alpha}{2}, n-1\right)}}\right) = 1 - \alpha$$

$$P\left(\frac{(n-1)s^2}{\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)}} > \sigma^2 > \frac{(n-1)s^2}{\chi^2_{\left(\frac{\alpha}{2}, n-1\right)}}\right) = 1 - \alpha$$

Which implies finally that $(1 - \alpha)100\%$ confidence interval for the population variance is given in the form

$$\frac{(n-1)s^2}{\chi^2_{\left(\frac{\alpha}{2}, n-1\right)}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)}}$$

- Example:

The following are the weight of **10** packages of grass seed distribution by a company: **46.4, 46.1, 45.8, 47, 46.1, 45.9, 45.8, 46.9, 45.2, 46**. Find 95% confidence interval for the variance.

Solution

$$\sum_{i=1}^{10} x_i = 461.2,$$

$$\sum_{i=1}^{10} x_i^2 = 21273.12$$

$$S^2 = \frac{1}{n-1} \left(\sum_{i=1}^{10} x_i^2 - \frac{(\sum_{i=1}^{10} x_i)^2}{n} \right) = \frac{1}{10-1} \left(21273.12 - \frac{(461.2)^2}{10} \right) = 0.25622$$

$$\alpha = 1 - 0.95 \rightarrow \alpha = 0.05 \rightarrow \frac{\alpha}{2} = 0.025$$

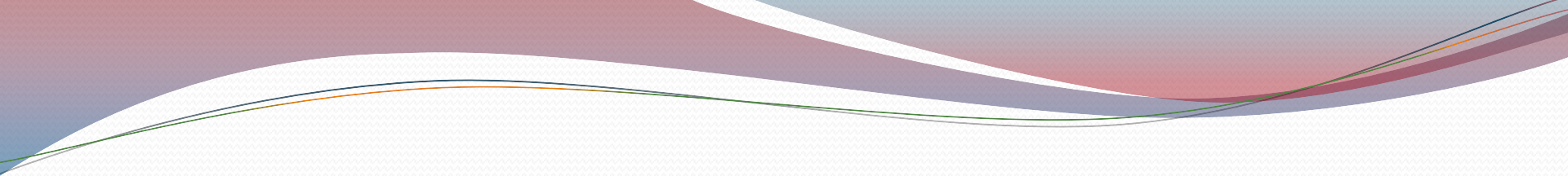
$$\chi^2_{\left(\frac{\alpha}{2}, n-1\right)} = \chi^2_{(0.025, 9)} = 19.023, \quad \chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)} = \chi^2_{(0.975, 9)} = 3.22$$

95% confidence interval for variance

$$\rightarrow \frac{(n-1)s^2}{\chi^2_{\left(\frac{\alpha}{2}, n-1\right)}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\left(1-\frac{\alpha}{2}, n-1\right)}}$$

$$\rightarrow \frac{(10-1) * 0.25622}{19.023} < \sigma^2 < \frac{(10-1) * 0.25622}{3.22}$$

$$\rightarrow 0.1354 < \sigma^2 < 0.799$$



Thank you for your
Attention!

